

Session 7: File Handling Topics: Streams (ifstream, ofstream), reading/writing files. Example: Save and retrieve creator analytics data.

Tasks

- 1) Create a text file called `my_fav_songs.txt` and write the names of your 5 favorite songs into it using `ofstream`.

Code-

```
#include<iostream>
#include<fstream>

using namespace std;

/*Create a text file called my_fav_songs.txt and write the names of your 5 favorite songs
into it using ofstream.*/
int main(){
    ofstream myfile("my_fav_songs.txt");

    myfile<<"\nbarbad";
    myfile<<"\nsikava nhi kishi se kisi se gila nhi ";
    myfile<<"\nadadt";
    myfile<<"\nGOat";
    myfile<<"\nyrr nhi milaya";

    myfile.close();

}
```

Output:

```
demo36.cpp  main6.cpp  demo36.cpp  demo36.cpp  demo36.cpp
my_fav_songs.txt
1
2  barbad
3  sikava nhi kishi se kisi se gila nhi
4  adadt
5  GOats
6  yrr nhi milaya

PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo36.cpp -o demo36 -static
>> .\demo36
PS C:\Users\Admin\OneDrive\Desktop\cpp> 
```

2) Read all song names from my_fav_songs.txt using ifstream and display each song on a new line in the console.

Code-

```
#include<iostream>
#include<fstream>
using namespace std;

/*Read all song names from my_fav_songs.txt using ifstream and display each song on a
new line in the console.
*/

int main(){

    ifstream fin;
    fin.open("my_fav_songs.txt");
    string data;
    while (getline(fin,data))

    {
        cout<<data<<endl; /* code */
    }
    fin.close();

    return 0;

}
```

output:

```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo37.cpp -o demo37 -static
>> .\demo37

barbad
sikava nhi kishi se kisi se gila nhi
adadt
GOat
yrr nhi milaya
PS C:\Users\Admin\OneDrive\Desktop\cpp> |
```

**3) Modify your code to append a new song name entered by the user to my_fav_songs.txt without overwriting the existing list.

Hint: Open the file in append mode using ofstream.**

Code-

```
#include<iostream>
#include<fstream>
#include<string>

using namespace std;

/*Modify your code to append a new song name eabntered by the user to
my_fav_songs.txt without
overwriting the existing list.<br><br><em><strong>Hint:</strong> Open the file in
append mode using ofstream.</em>*/
int main(){

    string songname;
    cout<<"enter new song ";
    getline(cin,songname);
    ofstream myfile("my_fav_songs.txt",ios::app);

    if (!myfile) {
        cout << "Error opening file!" << endl;
        return 1;
    }

    myfile<<songname<<endl;

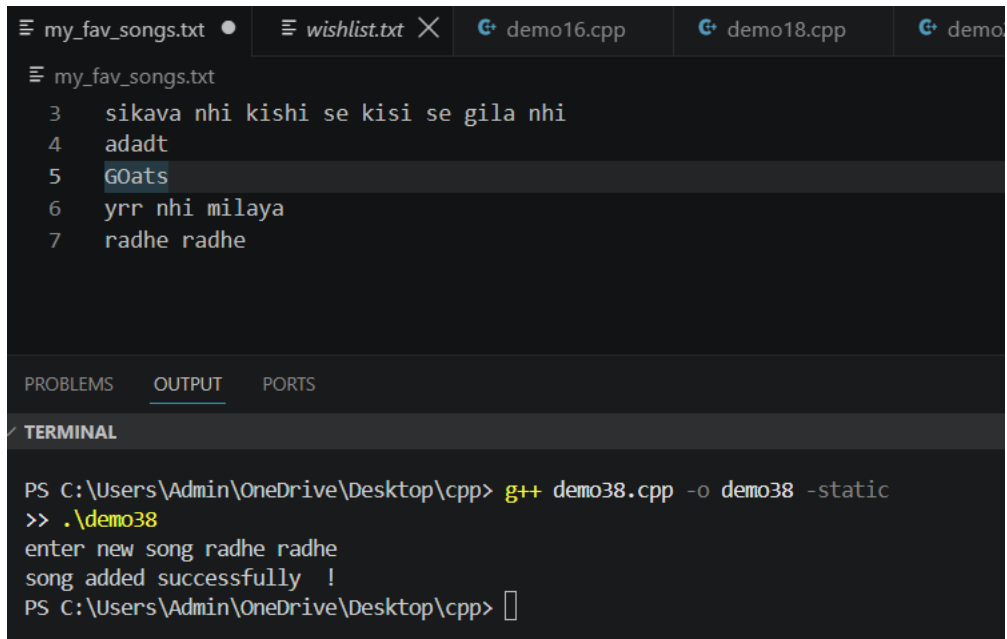
    myfile.close();

    cout<<"song added successfully !"<<endl;

    return 0;

}
```

Output:



The screenshot shows a code editor with several tabs: `my_fav_songs.txt`, `wishlist.txt`, `demo16.cpp`, `demo18.cpp`, and `demo`. The `my_fav_songs.txt` tab is active, displaying the following text:

```
3  sikava nhi kishi se kisi se gila nhi
4  adadt
5  GOats
6  yrr nhi milaya
7  radhe radhe
```

Below the editor, there is a **TERMINAL** panel. It shows the following commands and output:

```
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo38.cpp -o demo38 -static
>> .\demo38
enter new song radhe radhe
song added successfully !
PS C:\Users\Admin\OneDrive\Desktop\cpp> 
```

- 4) **Build a Flipkart-style wishlist tracker: ask the user to enter 3 product names and prices, save them to a file called wishlist.txt, then read the file and display each product with its price.**

Code-

```
#include<iostream>
#include<fstream>
#include<string>
using namespace std;

/*file handling
write
ofstream
read
*/

int main(){

    string productname;
    double price;

    ofstream file1("wishlist.txt");

    if(!file1){

        cout<<"error opening file !"<<endl;
        return 1;
    }

    for(int i= 1; i<= 3; i++)
    {
        cout<<"enter the product : "<<i<<"name:";
        getline(cin,productname);
```

```

        cout<<"enter price : "<<i<<"price";
        cin>>price;
        cin.ignore();

        file1<<productname<<" | "<<price<<endl;
    }

    file1.close();

    ifstream readfile1("wishlist.txt");
    if (!readfile1) {
        cout << "Error reading file!" << endl;
        return 1;
    }

    cout<<"\n===My Whishlist==="<<endl;
    string line;

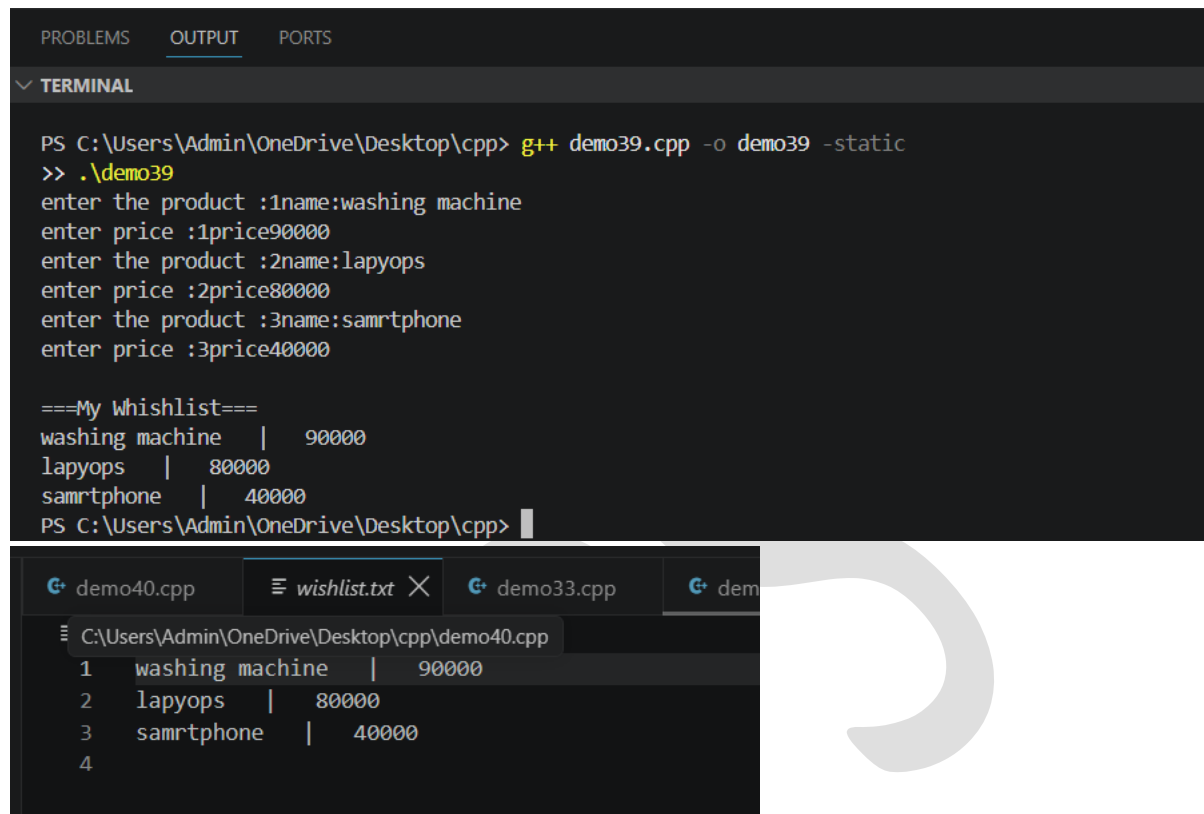
    while (getline(readfile1,line))
    {
        cout<<line<<endl; /* code */
    }
    readfile1.close();

    return 0;

}
/*Build a Flipkart-style wishlist tracker: ask the user to enter 3 product names and
prices,
save them to a file called wishlist.txt, then read the file and display each product with its
price.*/

```


output:



The image shows a screenshot of a C++ IDE. The top panel has tabs for 'PROBLEMS', 'OUTPUT', and 'PORTS', with 'OUTPUT' selected. Below this is a 'TERMINAL' tab. The terminal shows the execution of a C++ program. The user enters three products and their prices: 'washing machine' (90000), 'lapyops' (80000), and 'samrtphone' (40000). The program then displays a list titled '===My Whishlist==='. The bottom panel shows a code editor with a file named 'demo40.cpp' open. The code in the editor matches the output shown in the terminal.

```
PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ demo39.cpp -o demo39 -static
>> .\demo39
enter the product :1name:washing machine
enter price :1price90000
enter the product :2name:lapyops
enter price :2price80000
enter the product :3name:samrtphone
enter price :3price40000

===My Whishlist===
washing machine | 90000
lapyops | 80000
samrtphone | 40000
PS C:\Users\Admin\OneDrive\Desktop\cpp>
```

```
demo40.cpp wishlist.txt demo33.cpp dem
C:\Users\Admin\OneDrive\Desktop\cpp\demo40.cpp
1 washing machine | 90000
2 lapyops | 80000
3 samrtphone | 40000
4
```

5) Given a file called `insta_followers.txt` containing Instagram usernames (one per line), write a program to count and display the total number of followers listed in the file.

Constraint: Do not use any array or vector to store the names—just count as you read.

Code-

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    ifstream file("insta_followers.txt");

    if (!file) {
        cout << "Unable to open file!" << endl;
        return 1;
    }

    string username;
    int count = 0;

    while (getline(file, username)) {
        if (!username.empty()) {
            count++;
        }
    }

    file.close();

    cout << "Total number of followers: " << count << endl;

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\cpp> g++ de
>> .\demo40
Unable to open file!
PS C:\Users\Admin\OneDrive\Desktop\cpp> |
```